

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (R&AI) (CBCS) Degree Examinations

December 2023

RI2106-1 – DRIVE SYSTEMS FOR ROBOT

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **According to Pascal's Law, an increase in pressure at any point in an enclosed fluid is transmitted:**
 - A) In the opposite direction
 - B) In the perpendicular direction
 - C) In the same direction
 - D) In all directions
2. **In a hydraulic lift system, if a small force is applied to the smaller piston, what happens to the force exerted by the larger piston**
 - A) It decreases
 - B) It increases
 - C) It remains the same
 - D) It becomes zero
3. **What is the primary principle behind the functioning of a hydraulic system?**
 - A) Pascal's Law
 - B) Bernoulli's Principle
 - C) Archimedes' Principle
 - D) Ohm's Law
4. **The magnitude of the pressure of the fluid is not dependent on the**
 - A) Force
 - B) Area
 - C) Specific Weight
 - D) Temperature
5. **Gases are**
 - A) Compressible fluids
 - B) In-compressible fluids
 - C) Solid in phase
 - D) Super-compressive fluids
6. **If a hydraulic pump delivers 10 liters of fluid per minute, how much will it deliver in an hour?**
 - A) 600 liters
 - B) 300 liters
 - C) 900 liters
 - D) 120 liters
7. **How many cubic meters (m³) is equivalent to 1 liter (L)**
 - A) 0.001 m³
 - B) 0.01 m³
 - C) 0.1 m³
 - D) 1.0 m³
8. **In a gear pump, how many gears are typically used to move the fluid**
 - A) One
 - B) Two
 - C) Three
 - D) Four
9. **If a hydraulic pump delivers 8 liters of fluid per minute, how much will it deliver in 5 minutes**
 - A) 40 liters
 - B) 35 liters
 - C) 45 liters
 - D) 25 liters
10. **A hydraulic pump has an efficiency of 80%. If the pump takes in 100 liters of fluid, how much fluid is effectively pumped out**
 - A) 80 liters
 - B) 90 liters
 - C) 100 liters
 - D) 180 liters
11. **Which component of a PLC is responsible for processing and executing the program instructions**
 - A) Input module
 - B) CPU (Central Processing Unit)
 - C) Memory
 - D) Output module
12. **A hydraulic pump has an efficiency of 72.87%. What is the output power of the pump if the input power is 100 kW**
 - A) 72.87 kW
 - B) 159.06 kW
 - C) 37.13 kW
 - D) 6.29 kW

13. In a hydraulic motor, what type of energy is converted into mechanical energy
 - A) Hydraulic energy
 - B) Electrical energy
 - C) Thermal energy
 - D) Kinetic energy
14. In a pneumatic system, what is the purpose of an air reservoir
 - A) Store compressed air
 - B) Filter the compressed air
 - C) Regulate the air pressure
 - D) Generate compressed air
15. Which component is responsible for maintaining a specific pressure level in a pneumatic system
 - A) Air filter
 - B) Pressure regulator
 - C) Air cylinder
 - D) Air compressor
16. Which component is used to control the direction of airflow in a pneumatic system
 - A) Directional control valve
 - B) Pressure gauge
 - C) Air compressor
 - D) Air reservoir
17. In electro-pneumatics, what component is used to convert electrical signals into pneumatic actions
 - A) Pressure regulator
 - B) Solenoid valve
 - C) Pneumatic cylinder
 - D) Air compressor
18. The main purpose of an electro-pneumatic system is to
 - A) Control the flow of compressed air using electrical signals
 - B) Convert pneumatic energy into Mechanical energy
 - C) Convert electrical energy into pneumatic energy
 - D) Amplify the force of compressed air
19. What is the typical operating frequency range of a VFD
 - A) 10 Hz - 60 Hz
 - B) 20 Hz - 120 Hz
 - C) 30 Hz - 180 Hz
 - D) 40 Hz - 240 Hz
20. What type of load is a variable frequency drive typically used for
 - A) Resistive loads
 - B) Inductive loads
 - C) Capacitive loads
 - D) Impedance Load

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive an equation for Volumetric flow rate, Energy, Power from governing laws used in fluid power system.	06	L1	1	1
	b) In a hydraulic jack, a force of 100 N is exerted on the small piston. Determine the upward force on the larger piston. The area of the smaller piston is 50 cm ² and the area of the larger piston is 500 cm ² . If the small piston moves 10 cm, how far will the large piston move? Assume the oil is incompressible.	04	L2	1	1
	c) Explain with a neat sketch the working principle of balanced type vane pump.	06	L1	2	1
2.	a) Define the following with respect to Hydraulic Pumps: i) Overall Efficiency ii) Mechanical Efficiency iii) Volumetric Efficiency	06	L1	1	1
	b) Write the Ladder diagram for the following Logic Gates: i) AND ii) OR iii) NAND iv) NOR v) EX-OR	10	L1	2	1
3.	a) What is the difference between timer and counter in PLC.	05	L1	2	1
	b) Explain the concept of Latching using PLC with an example.	05	L1	2	1
	c) Double acting cylinder is used to perform to and fro operation. Cylinder has to move forward when PB1 button is pressed and continue to and fro motion till 10 cycles of operations is performed. Draw the pneumatic circuit, PLC wiring diagram and ladder diagram to implement this task.	06	L2	2	1

Unit – II

4. a) A hydraulic motor has a displacement of 150 cm³ and operates with a pressure of 120 bars and a speed of 2500 rpm. The actual flow rate consumed by the motor is 0.0078 m³/s and the actual torque delivered by the motor is 250 Nm. Find:
- i) Volumetric efficiency
 - ii) Mechanical efficiency
 - iii) Overall efficiency
 - iv) Power delivered by the motor.
- b) Differentiate between Hydraulic System and Pneumatic System.
5. a) Explain the working of pressure regulator with a neat sketch.
- b) Using a rotary indexing table plastic containers are to be separated in linear sequence. By pressing a pushbutton switch the oscillating piston rod of a cylinder drives the rotary table in sequence via a pawl. When the pushbutton is pressed again, this drive is switched off. Explain the electro pneumatic circuit.
6. a) A hydraulic motor has a 100cm³ volumetric displacement. If it has a pressure rating of 140 bar and receives oil from a 0.001 m³ theoretical flow rate pump, find the motor (i) speed, (ii) theoretical torque, (iii) theoretical kW power.
- b) Using a diverting device parts are to be removed from one conveyor track onto another in linear sequence. By pressing a pushbutton switch the oscillating piston rod of a cylinder pushes the turntable via a pawl in stepped sequence. The parts are diverted and transported onwards in the opposite direction. By pressing another pushbutton switch the drive unit is switched off. Explain the electro pneumatic circuit.

08	L3	3	1
08	L2	3	1
08	L2	4	1
08	L2	4	1
08	L2	3	1
08	L2	4	1

Unit – III

7. a) What is the difference between harmonic drives and variable frequency drive?
- b) What are the different types of specialized electric drives?
- c) What are the advantages and disadvantages of variable frequency drive?
8. a) With a neat sketch explain the working principle of servo drive.
- b) With a neat sketch explain the working principle of variable frequency drive.

08	L1	5	1
04	L1	5	1
04	L1	5	1
08	L1	5	1
08	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
